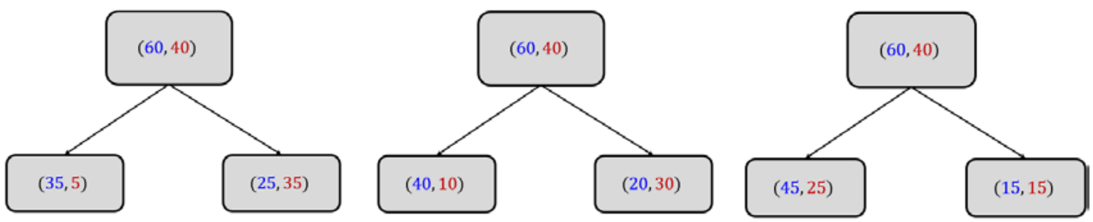
HW5

- **Submission date: 03/17/2025 (11:59 PM)**
- Read your class notes *thoroughly* before attempting the assignment.
- You can submit ***well commented*** code in either `.py` or `.ipynb` formats.
- Submitting a PDF explaining your code is *highly* encouraged.

- The Scikit-Learn assignment for Decision Trees has been deferred to next week since the discussion on the topic (including ID3) is incomplete as of now.
- For the following questions, you may use calculators, but not programming (use NumPy/Python only for verification. Do not submit that code).
- **Textbook for this homework:** Lipschutz, S. and Lipson, M., 2001. *Schaum's outline of theory and problems of Probability (3rd edition)*. An online copy is available on BU's library website. This book has been reserved for this course. If you cannot access it, please email Tracy Stone (tstone@bu.edu), or Sarah Goodwin (circulation@bu.libanswers.com) for help.

1. **Impurity Metrics**. Compute the change in (a) misclassification rate, (b) Gini index, and (c) Entropy for each of the following splits (full points will be given only if all intermediate steps are clearly shown).



2. **Gini Index**. Suppose we have a classification problem with $\mathcal{C}$ classes and $\widehat{p_i}$ is the proportion of Class i $(1 \leq i \leq \mathcal{C})$ in a certain tree node.
   Can the Gini index for this node be written as $L_{Gini} = 1 - \sum_{i=1}^{\mathcal{C}} \widehat{p_i}^2$? If yes, how does this expression follow from the one discussed in class, and if not, why not?

3. **Variance and Standard Deviation**. Please read Example 6.8 in the textbook before answering this question. Answer questions 6.71 and 6.72.

4. **Covariance and Correlation Coefficient**. Answer questions 5.77 (parts (a) and (c)) and 5.91.